Implementation Workflow

> **Document Owner:** Platform Team | **Status:** Approved | **Last Updated:** February 2026

Overview

This document provides step-by-step workflows for implementing Golden Rules in new data domains and conducting compliance audits.

New Domain Implementation (8 Weeks)

Phase 1: Planning & Design (Week 1)

Step	Deliverable	Owner
1.1 Requirements gathering	Requirements document	Data Steward
1.2 Data classification	Classification matrix	Data Steward + Security
1.3 Architecture design	ERD, data flow diagram	Data Engineer
1.4 Schema definition	Gold layer YAML files	Data Engineer

Checklist:

- [ ] Business requirements documented
- [ ] Data sources identified
- [ ] PII columns identified and classified
- [ ] Gold layer star schema designed (ERD)
- [ ] YAML schema files created

Phase 2: Infrastructure Setup (Week 2)

Step	Deliverable	Owner
2.1 Unity Catalog setup	Catalog and schemas	Platform Engineer
2.2 Asset Bundle creation	Bundle directory	Platform Engineer
2.3 Access control	Permissions granted	Platform Admin

Checklist:

- [ ] Catalog created (if new)
- [ ] Schemas created (bronze, silver, gold)
- [ ] Predictive optimization enabled
- [ ] Asset Bundle validated
- [ ] Service principal configured

Phase 3: Bronze Layer (Week 3)

Step	Deliverable	Owner
3.1 Create Bronze tables	DDL with metadata	Data Engineer
3.2 Build ingestion pipeline	Auto Loader config	Data Engineer
3.3 Deploy Bronze job	Working ingestion	Data Engineer

Checklist:

- [ ] Tables created with `USING DELTA CLUSTER BY AUTO`
- [ ] CDF enabled (`delta.enableChangeDataFeed = true`)
- [ ] Table and column COMMENTs added
- [ ] Ingestion metadata columns present
- [ ] Job running successfully

Phase 4: Silver Layer (Week 4)

Step	Deliverable	Owner
4.1 Design DLT pipeline	Notebook structure	Data Engineer
4.2 Implement data quality	Expectations (5+ per table)	Data Engineer
4.3 Deploy DLT pipeline	Working pipeline	Data Engineer

Checklist:

- [ ] DLT pipeline uses Direct Publishing Mode
- [ ] All tables have `cluster_by_auto=True`
- [ ] Critical rules use `expect_or_drop`
- [ ] Quarantine tables capture failures
- [ ] Pipeline running with serverless + Photon

Phase 5: Gold Layer (Week 5)

Step	Deliverable	Owner
5.1 Create Gold tables	Tables from YAML	Data Engineer
5.2 Apply constraints	PKs and FKs	Data Engineer
5.3 Build MERGE pipelines	Merge scripts	Data Engineer
5.4 Deploy Gold jobs	Setup + merge jobs	Data Engineer

Checklist:

- [ ] Tables created from YAML schema
- [ ] All dimensions have PRIMARY KEY

- [ ] All facts have composite PRIMARY KEY
- [ ] FKs applied AFTER all PKs exist
- [ ] MERGE uses deduplication pattern
- [ ] SCD Type 2 for historical dimensions

Phase 6: Semantic Layer (Week 6)

Step	Deliverable	Owner
6.1 Create TVFs	SQL functions	Analytics Engineer
6.2 Create Metric Views	YAML definitions	Analytics Engineer
6.3 Configure Genie Space	Trusted assets	Analytics Engineer

Checklist:

- [ ] TVFs use STRING date parameters
- [ ] Metric Views use v1.1 format (no `name` field)
- [ ] Comments follow v3.0 structured format
- [ ] No transitive joins in Metric Views
- [ ] Genie Space tested with sample questions

Phase 7-8: Production (Weeks 7-8)

Step	Deliverable	Owner
7.1 Setup Lakehouse Monitoring	Monitors configured	Data Engineer
7.2 Create alerts	Alert rules	Data Engineer
7.3 Production deployment	Production jobs	Platform Engineer
7.4 Documentation	Runbook	Data Engineer

Checklist:

- [ ] Monitors created for Gold tables
- [ ] Custom metrics documented
- [ ] Freshness alerts configured
- [ ] Quality threshold alerts configured
- [ ] Production schedules enabled
- [ ] Runbook completed

Compliance Audit Workflow

Step 1: Inventory

```sql
-- List all tables by layer
SELECT
    table_catalog,
    table_schema,
    table_name,
    CASE
        WHEN table_schema LIKE '%bronze%' THEN 'Bronze'
        WHEN table_schema LIKE '%silver%' THEN 'Silver'
        WHEN table_schema LIKE '%gold%' THEN 'Gold'
    END as layer
FROM information_schema.tables
WHERE table_catalog = 'my_catalog';
```

Step 2: Rule Assessment

Check CDF Enabled (Bronze/Silver):

```sql
SELECT
    table_schema, table_name,
    CASE WHEN option_value = 'true' THEN '✅' ELSE '❌' END as cdf_enabled
FROM information_schema.tables t
LEFT JOIN information_schema.table_options o
    ON t.table_name = o.table_name
    AND o.option_name = 'delta.enableChangeDataFeed'
WHERE t.table_schema IN ('bronze', 'silver');
```

Check PK Coverage (Gold):

```sql
SELECT
    table_name,
    CASE WHEN COUNT(constraint_name) > 0 THEN '✅' ELSE '❌' END as has_pk
FROM information_schema.table_constraints
WHERE constraint_type = 'PRIMARY KEY' AND table_schema = 'gold'
GROUP BY table_name;
```

Check Documentation (All):

```sql
SELECT
    table_name,
    CASE WHEN comment IS NOT NULL AND LENGTH(comment) > 50 THEN '✅' ELSE
'❌' END as documented
FROM information_schema.tables
WHERE table_schema IN ('bronze', 'silver', 'gold');
```

Step 3: Findings Report

Severity	Action
Critical	Immediate remediation required
High	Remediation within 1 sprint
Medium	Add to backlog
Low	Track for next audit

Exception Request Process

Step 1: Submit Request

Complete the Exception Request Form with:

- Rule being bypassed
- Business justification
- Risk assessment
- Remediation timeline

Step 2: Review

Risk Level	Approver
Low/Medium	Platform Architect
High	Architecture Review Board

Step 3: Document & Track

- Approved exceptions logged in governance system
- Review date scheduled
- Conditions documented

Timeline Summary

Phase	Duration	Key Milestone
Planning	1 week	YAML schemas complete
Infrastructure	1 week	Asset Bundle deployed
Bronze	1 week	Ingestion running
Silver	1 week	DLT pipeline running
Gold	1 week	MERGE jobs running
Semantic	1 week	Metric Views deployed
Production	2 weeks	Monitoring active

Total: 8 weeks for complete domain implementation

Related Documents

- Data Governance
- Roles & Responsibilities
- Data Modeling
- Naming & Comment Standards
- Tagging Standards
- Data Quality Standards
- Architecture Review Checklist